

NexusAI

LLM API Fallback Routing and Logs Tracing Gateway for AI Model Providers

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1. Problem Definition

NexusAI Gateway solves a common problem faced by anyone building on top of large language model APIs: providers such as Google Gemini, Groq, OpenRouter, NVIDIA NIM and Cerebras etc enforce rate limits, and once a limit is hit an AI application can stop working or need a manual provider switch. NexusAI provides one unified API endpoint that automatically falls over to another configured provider whenever a retryable failure occurs — whether that provider is being used on a free tier or a paid plan. This makes it equally useful for students experimenting on free-tier keys and for working professionals running production workloads on paid plans who simply want their application to keep running smoothly.

2. Key Features

- Unified API endpoint for multiple LLM providers.
- Automatic provider fallback on rate limits or failures.
- Works seamlessly across free-tier and paid provider plans.
- Five supported AI providers out of the box.
- JWT authentication.
- Request logging.
- Provider health monitoring.
- Spring Boot microservices architecture.
- LLM Request and Response Logging
- Observability On Api Calls
- User Friendly Interface
- Easy To Configure & Manage Different LLM APIs

3. Technology Stack

Component	Details
Backend	Spring Boot 3.5 (Java 17)
Frontend	React
Microservices	Spring Cloud Gateway & Eureka
Security	Spring Security + JWT
Database	MySQL
Build Tool	Maven
Containerization	Docker
Testing	Postman
LLM Providers	Gemini, Groq, OpenRouter, NVIDIA NIM, Cerebras and more

4. Stakeholders

- **AI Developers:** Use one API instead of managing multiple providers, whether on free or paid plans.
- **Students:** Learn Spring Boot microservices through a practical infrastructure project.
- **Working Professionals:** Rely on automatic failover to keep production applications running smoothly.
- **Development Teams:** Reduce downtime caused by provider rate limits or outages.
- **Mentors & Faculty:** Evaluate a real-world backend architecture project.

5. Expected Outcomes

- Single integration point for multiple LLM providers.
- Automatic failover during rate limits, on both free and paid plans.
- Interactive and Simple User Friendly UI To Manage LLMs API
- Simplified API management.
- Practical understanding of Gateway, Eureka, JWT, Docker and REST communication.
- Portfolio-ready backend infrastructure project.